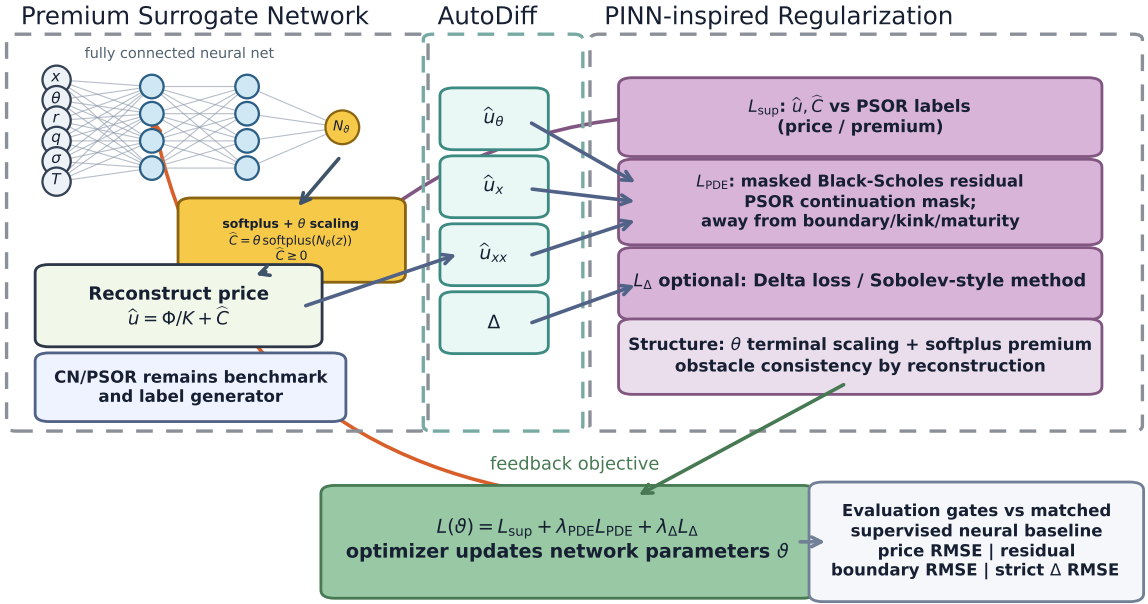Project-specific PINN-inspired mechanism